

Tokenization

How Text Becomes Numbers in NLP

Tokenization ■ Vocab ■ Embeddings ■ Context Windows

Motivation

Why do LLMs need numbers?

- ▶ Neural networks operate only on numbers
- ▶ Text is symbolic
- ▶ We must convert:

Text → Tokens → Token IDs → Embeddings → Tensors

This is the foundation of all NLP and LLMs.